

HEART DISEASE PREDICTION USING MACHINE LEARNING

Abstract— With heart disease standing as a predominant cause of global mortality, the imperative for advanced diagnostic and predictive methodologies has never been more apparent. Machine Learning (ML) emerges as a pivotal tool, promising a paradigm shift in the landscape of cardiovascular health interventions. This research, under the title "Heart Disease Prediction using Machine Learning" seeks to harness the potential of ML in scrutinizing intricate health parameters for predicting occurrences of heart disease. The research is framed by two principal aims. Firstly, it endeavors to develop a robust machine learning-based predictive model, drawing insights from the Heart Disease UCI dataset. The envisioned model holds the promise of delivering timely alerts, thereby mitigating the risk of severe cardiac events and contributing to improved patient outcomes. Secondly, the study comprehensively explores various ML algorithms, encompassing Logistic Regression, Decision Trees, Random Forests, Gradient Boosting Machines, Neural Networks, and Support Vector Machines, evaluating their efficacy in the nuanced realm of heart disease prediction.

Keywords— *Machine learning, Logistic Regression, Decision Trees, Random Forests, Neural Networks, Support Vector Machines*

I. INTRODUCTION

The global prevalence of cardiovascular diseases underscores the critical need for advancements in diagnostic and predictive methodologies. In response, Machine Learning (ML) has emerged as a transformative tool, presenting unprecedented opportunities for reshaping cardiovascular health interventions. This study, aims to contribute significantly to the field by leveraging ML capabilities. Our primary objective is the development of a robust predictive model for the early detection of heart diseases, utilizing insights gleaned from the comprehensive Heart Disease UCI dataset. This model holds promise in providing timely alerts, thereby reducing the risk of severe cardiac events and enhancing patient outcomes. The research commences with a comprehensive exploration of various ML algorithms, including Logistic Regression, Decision Trees, Random Forests, Gradient Boosting Machines, Neural

Networks, and Support Vector Machines. This exploration seeks to discern the efficacy of each algorithm in the intricate domain of heart disease prediction, forming the basis for informed model development and selection.

Introducing an additional layer of complexity, the study incorporates ensemble techniques to integrate predictions from diverse models. This strategic integration aims to enhance the accuracy and robustness of our predictive model, ensuring effectiveness across diverse scenarios. Evaluation metrics for the developed models encompass accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Moreover, the study underscores the critical importance of model interpretability, particularly in a healthcare context. The goal is to ensure that the final model is not only accurate but also transparent, empowering healthcare professionals to comprehend and trust the rationale behind its predictions (Bhajibhakare et al., 2019; Abdalrada et al., 2022).

II. OVERVIEW OF DATASET WITH HYPOTHESIS

A. Heart Disease UCI

Link to the dataset:

<https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data?resource=download>

The dataset under consideration is multivariate, encompassing diverse mathematical and statistical variables relevant to cardiovascular health. Comprising 14 attributes such as age, sex, chest pain type, and others, the dataset originated from the Cleveland database, widely utilized by machine learning researchers. Despite initially consisting of 76 attributes, studies have predominantly focused on a subset of 14 attributes. The primary objective of analyses on this dataset involves predicting the presence of heart disease based on patient attributes. We aim to conduct experimental tasks, extract insights to enhance understanding and diagnosis of cardiovascular issues.

Column Name	Description	Data Type
id	Unique id for each patient	Integer
age	Age of the patient in years	Integer
sex	Male/Female	String (Object)
dataset	Place of study (origin)	String (Object)
cp	Chest pain type (typical angina, atypical angina, non-anginal, asymptomatic)	String (Object)
trestbps	Resting blood pressure (in mm Hg on admission to the hospital)	Float
chol	Serum cholesterol in mg/dl	Float
fb	If fasting blood sugar > 120 mg/dl	Boolean
restecg	Resting electrocardiographic results (normal, stt abnormality, lv hypertrophy)	String (Object)
thalach	Maximum heart rate achieved	Float
exang	Exercise-induced angina (True/ False)	Boolean
oldpeak	ST depression induced by exercise relative to rest	Float
slope	The slope of the peak exercise ST segment	String (Object)
ca	Number of major vessels (0-3) colored by fluoroscopy	Float
thal	Thalassemia types (normal; fixed defect; reversible defect)	String (Object)
num	The predicted attribute	Integer

B. Hypothesis

A machine learning model, trained on a comprehensive dataset including attributes such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, oldpeak (ST depression induced by exercise relative to rest), the slope of the peak exercise ST segment, number of major vessels, and Thalassemia, will be able to accurately predict the presence or absence of heart disease in patients. The hypothesis assumes that these features contain discernible patterns and relationships that the machine learning algorithm can learn and generalize to new, unseen data. Successful validation of this hypothesis would contribute to the development of an effective predictive model for heart disease diagnosis.

III. LITERATURE REVIEW

Heart disease remains one of the foremost health challenges worldwide. Its early detection and accurate prediction play a pivotal role in effective treatment and management. With the rapid advancements in technology and data science, machine learning (ML) techniques are increasingly being employed to enhance the accuracy and efficiency of heart disease prediction. This review delves into some of the recent studies that have utilized ML for heart disease prediction.

a) Machine Learning and Coronary Heart Disease in Gout Patients

Jiang et al. (2021) embarked on a noteworthy exploration of predicting coronary heart disease in gout patients using machine learning models. Gout, a form of inflammatory arthritis, has long been associated with an increased risk of coronary heart disease. However, predicting which gout patients are at heightened risk remains a challenge. By employing machine learning models, Jiang and her colleagues provided a more nuanced and data-driven approach to this problem. Their work underscores the potential of machine learning in targeting specific patient populations for more accurate disease prediction (Jiang et al., 2021).

b) Predicting Cardiovascular Disease: The Role of Dataset Augmentation

In another study, Ahmed et al. (2022) highlighted the importance of dataset augmentation in predicting cardiovascular disease. They used multiple machine learning models on self-augmented datasets of heart patients. Dataset augmentation, which involves artificially expanding the dataset through techniques such as oversampling, can be especially crucial when dealing with imbalanced datasets. The researchers' efforts spotlight the necessity of not just choosing the right ML model but also ensuring that the underlying data is robust and representative (Ahmed et al., 2022).

c) The Interplay between Elastin, Arterial Mechanics, and cardiovascular disease

Wagenseil et al. (2018) delved into a more mechanistic understanding of heart disease by exploring the role of elastin and arterial mechanics. While this study might not be directly tied to machine learning predictions, understanding the underlying mechanics and molecular players in heart disease can offer invaluable insights for feature selection in ML models. Their research sheds light on the fundamental biological processes that can be potential predictors or features in ML-driven heart disease prediction models (Wagenseil et al., 2018).

d) Advancements in ML Models for Cardiovascular Disease Prediction

A team led by Yarasuri (2019) focused on the development of machine learning models for cardiovascular disease prediction. Their study emphasized the relevance of combining clinical and genetic data to enhance predictive accuracy. The integration of genetic data points towards the future of personalized medicine, where treatment and preventive measures can be tailored to an individual's unique genetic makeup. This multi-faceted approach, combining clinical data, genetic insights, and advanced ML algorithms, is indicative of the direction in which heart disease prediction is moving (Yarasuri et al., 2019).

e) ML in Intensive Care Units: Discharge Prediction for Cardiovascular Diseases Patients

In a study conducted by Karboub and Tabaa (2022), the researchers applied machine learning to predict the discharge of cardiovascular disease patients in intensive care units (ICUs). Given the dynamic and critical nature of ICUs, timely and accurate predictions can greatly aid in patient management and resource allocation. Their study underscores the versatility of machine learning, which can be applied not just for disease prediction but also for patient management in critical care settings (Karboub & Tabaa, 2022).

f) Algorithmic Approaches: KNN and RTC in Heart Disease Prediction

G. Senthil et al. (2022) explored heart disease prediction using the K-nearest neighbors (KNN) and Random Tree Classifier (RTC) algorithms. Their research emphasized the significance of data preprocessing, feature selection, and algorithm optimization. The comparative analysis of KNN and RTC offers insights into the strengths and limitations of different algorithms, guiding researchers, and practitioners in selecting the most suitable algorithm for specific datasets and prediction challenges (G. Senthil et al., 2022.).

Machine learning offers a potent set of tools for the prediction and management of heart disease. The studies reviewed here showcase the versatility and depth of ML in addressing various facets of heart disease prediction, from understanding the underlying biology to enhancing dataset robustness and algorithmic accuracy. As technology continues to evolve, it's evident that ML will play an increasingly central role in revolutionizing the way we predict and treat heart disease.

IV. PROBLEM DESCRIPTION

This study delves into the intricacies of developing a robust machine learning-based predictive model for heart disease detection, utilizing insights derived from the Heart Disease UCI dataset. While the dataset offers rich information, challenges arise in preprocessing, including handling missing values, encoding categorical variables, and normalizing numerical attributes. These preprocessing hurdles require meticulous consideration to ensure data integrity and suitability for subsequent machine learning algorithms.

A critical aspect of the research involves a comprehensive exploration of various machine learning algorithms, ranging from Logistic Regression to Support Vector Machines. The goal is to discern the efficacy of each algorithm in the nuanced realm of heart disease prediction, guiding the selection of the most suitable models that balance predictive accuracy and computational efficiency. The effectiveness of the developed models is rigorously evaluated using a suite of metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. This comprehensive evaluation ensures a nuanced understanding of model performance under various

conditions, guiding the selection of models that align with the goals of accurate and reliable heart disease prediction.

The research places a significant emphasis on model interpretability, acknowledging its critical importance in healthcare contexts. Ensuring that the final model is not only accurate but also transparent is essential for fostering trust among healthcare professionals, enabling them to comprehend and act upon the rationale behind predictions.

V. RESEARCH AIMS AND OBJECTIVES

a) RESEARCH AIMS

- Aim 1: To develop a machine learning-based predictive model for heart disease detection using the Heart Disease UCI dataset.
- Aim 2: To enhance early interventions in cardiovascular health through timely and accurate not differentiated among departments of the same organization).

b) Research Objectives

- Objective 1: Understand and preprocess the Heart Disease UCI dataset to make it suitable for machine learning algorithms.
- Objective 2: Investigate various machine learning algorithms, including Logistic Regression, Decision Trees, Random Forests, Gradient Boosting Machines, Neural Networks, and Support Vector Machines, to determine their efficacy in predicting heart disease.
- Objective 3: Implement ensemble techniques to combine predictions from multiple models, aiming to improve accuracy and robustness.
- Objective 4: Evaluate the developed models using appropriate metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.
- Objective 5: Ensure the interpretability of the final model, allowing healthcare professionals to understand the basis of its predictions

c) Research Questions

- RQ1: How effective are various machine learning algorithms in predicting heart disease using the Heart Disease UCI dataset?
- RQ2: What preprocessing techniques, including handling missing values, encoding categorical variables, and normalizing numerical attributes, are most effective for this dataset?
- RQ3: How does the implementation of ensemble techniques impact the accuracy and robustness of heart disease predictions?

- RQ4: Which performance metrics provide the most insightful evaluation of the developed models in the context of heart disease prediction?
- RQ5: How can the interpretability of the developed model be ensured, and what insights can be derived from its predictions?

VI. PROPOSED METHODOLOGY

A. Data Understanding

- **Total Entries and Features:** The dataset comprises 920 entries (rows) and 16 features (columns).
- **Feature Details:** The dataset includes a mix of numerical and categorical variables.
- The features are id, age, sex, dataset, cp (chest pain type), trestbps (resting blood pressure), chol (serum cholesterol), fbs (fasting blood sugar), restecg (resting electrocardiographic results), thalch (maximum heart rate achieved), exang (exercise induced angina), oldpeak, slope, ca (number of major vessels colored by fluoroscopy), thal, and num (diagnosis of heart disease).
- **Non-Null Counts:** All columns have non-null entries, although some features like ca and thal have a significant number of missing values.
- **Diagnosis of Heart Disease (num):** Scores range from 0 to 4, likely indicating the severity or presence of heart disease.

B. Descriptive Statistics

- **Age:** Patients' ages range from 28 to 77 years, with an average age of approximately 53.5 years.
- **Resting Blood Pressure (trestbps):** Ranges from 0 to 200 mm Hg, with an average around 132 mm Hg.
- **Serum Cholesterol (chol):** Varies widely, from 0 to 603 mg/dl, indicating a diverse patient profile in terms of cholesterol levels.
- **Maximum Heart Rate Achieved (thalch):** Ranges from 60 to 202 beats per minute.
- **Number of Major Vessels (ca):** Ranges from 0 to 3.

C. Exploratory Data Analysis

The Exploratory Data Analysis (EDA) conducted on the heart disease dataset provides several key insights:

Fig 1 shows histograms of numerical features, we can infer the following from the graph.

Observations:

- **Age:** Most individuals are in the 50-60 age range.

- **trestbps (Resting Blood Pressure):** Concentrated around 120-140 mm Hg, indicating a common range for resting blood pressure in the dataset.
- **chol (Serum Cholesterol):** Majority of the values are clustered between 200 and 300 mg/dl.
- **thalch (Maximum Heart Rate Achieved):** Shows a wide range of values, mostly between 100 and 160 beats per minute.
- **oldpeak:** Most values are close to 0, with a few extending up to 6.
- **ca (Number of Major Vessels):** Concentrated at 0, indicating most individuals have no major vessels colored by fluoroscopy.
- **num (Heart Disease Diagnosis):** A widespread across different diagnosis categories, with a notable number of cases in the 0 (no heart disease) and 1 category.

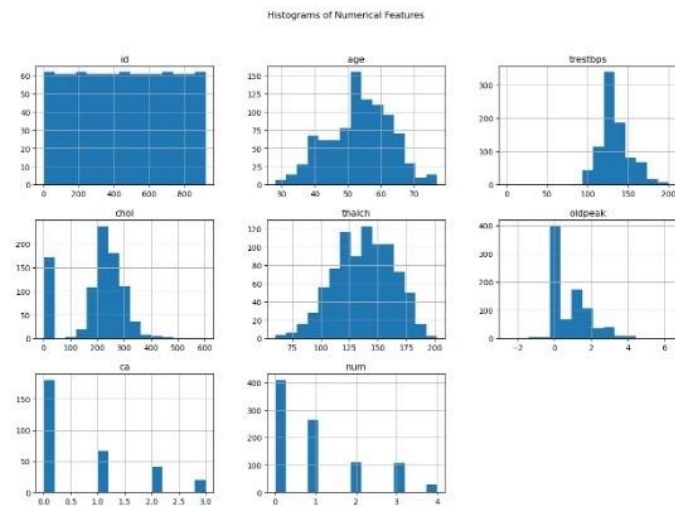


FIG 1: HISTOGRAMS OF NUMERICAL FEATURES

Fig 2 shows boxplots of numerical features. We have used boxplots mainly to identify the central tendency and spread of the data, along with outliers.

Observations:

- **Outliers:** Notably present in trestbps, chol, and oldpeak. These outliers may represent extreme but valid variations or could be data entry errors.
- **Median and IQR:** For features like age, trestbps, and chol, the boxplots show the median (central line in the box) and the interquartile range (IQR, length of the box), indicating the spread of most of the data.

Fig 3 shows correlation heatmap of numerical features. We have utilized heatmap to assess the linear relationships between numerical variables.

Observations:

- **Strong Correlations:** Not evident in this dataset, as most values are below 0.5 in absolute value.
- **Notable Correlations:** Slight correlations may be observed, for example, between age and trestbps, but

they are not strong enough to indicate a significant linear relationship.

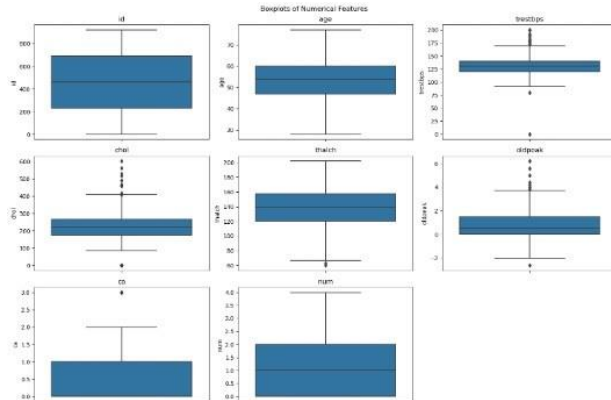


FIG 2: BOXPLOTS OF NUMERICAL FEATURES

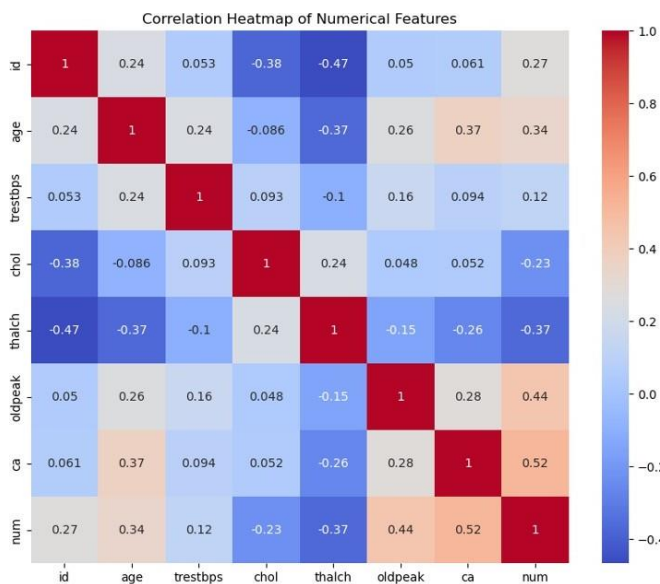


FIG 3: CORRELATION HEATMAP OF NUMERICAL FEATURES

The pair plot is a matrix of plots used to visualize the relationship between multiple numerical variables. In this specific analysis, the selected features are age, trestbps (resting blood pressure), chol (serum cholesterol), thalch (maximum heart rate achieved), and oldpeak. The hue, which differentiates the data points in each plot, is determined by the num variable, representing the diagnosis of heart disease.

Diagonal Plots (Kernel Density Estimates - KDEs):

- Purpose: KDEs are used to visualize the probability density distribution of a continuous variable.
- Features' Distributions:
- Each diagonal plot in the pair plot matrix is a KDE for one of the selected variables.

- These plots show the distribution of each variable segmented by the heart disease diagnosis categories (as indicated by different hues/colors).

- For example, the KDE for age will show how age distribution varies across different heart disease categories. A peak in the KDE plot for a particular age range in a specific heart disease category would indicate a higher density of patients in that age range with that category of heart disease.

D. Data Pre-processing

Handling Missing Values

- Numerical Columns: Missing values in numerical columns were filled with the median of their respective columns. This approach is robust to outliers and helps maintain the central tendency of the data without introducing bias.

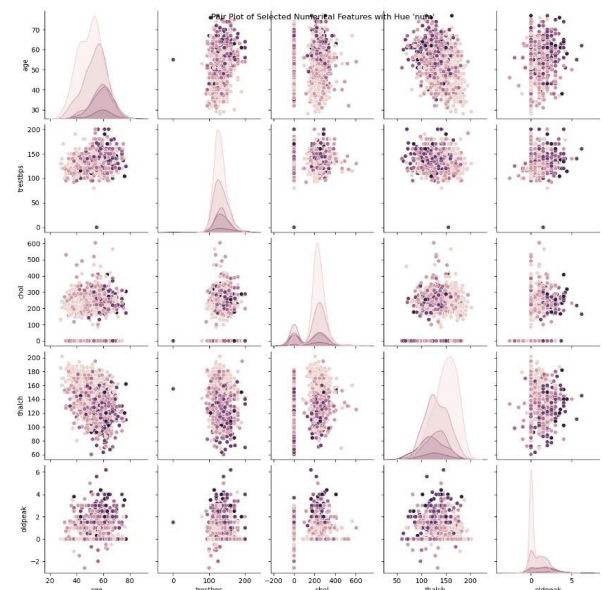


FIG 4: PAIR-WISE PLOT

- Categorical Columns: Missing values in categorical columns were filled with the mode (the most frequent value) of their respective columns. This is a standard practice for imputing missing categorical data, ensuring the most common category is used for replacement.

Encoding Categorical Variables

- One-Hot Encoding: Categorical variables were transformed using one-hot encoding. This process creates binary (0/1) columns for each category of a categorical variable, which is essential for modeling as it allows algorithms to better interpret these features.

Normalizing Numerical Attributes

- **MinMaxScaler:** The MinMaxScaler was used to normalize the numerical columns, scaling each feature to a given range, typically between 0 and 1. This normalization is crucial for models that are sensitive to the scale of the data, like distance-based algorithms or neural networks.

Transforming the Target Variable

- **Binary Classification:** The 'num' variable, representing heart disease diagnosis, was converted into a binary classification problem (0 for no disease, 1 for any disease presence). This transformation simplifies the analysis and is suitable for binary classification models.

E. Feature Selection

Correlation Analysis

- **Correlation Matrix:** The correlation matrix for numerical columns was computed. This matrix measures the linear relationships between each pair of numerical features, including the target variable num.
- **Correlation with Target Variable:** The correlation of each feature with the target variable num was extracted and sorted in descending order.

Top Feature Selection

- **Top Features:** The top 10 features (excluding the target variable itself) were selected based on their correlation with num. These features are likely to have a stronger linear relationship with the heart disease diagnosis, making them potentially more informative for predictive modeling.
- **Selected Features:** The top features include oldpeak, id, age, ca, trestbps, chol, and thalch, among others.

F. Model Development

Logistic Regression

- Accuracy: 81.52%
- F1 Score: 83.96%
- Recall: 81.65%
- ROC-AUC: 83.79%

Logistic Regression performs reliably with good accuracy and a balanced F1 score. The ROC-AUC score indicates a strong ability to distinguish between the classes. High recall implies it's effective in identifying true positive cases of heart disease.

Random Forest

- Accuracy: 85.33%
- F1 Score: 87.20%
- Recall: 84.40%
- ROC-AUC: 91.09%

Random Forest model shows superior performance in all metrics compared to Logistic Regression, particularly in ROC-AUC, suggesting excellent classification capability. This model balances accuracy with the ability to detect positive cases well.

SVM (Support Vector Machine)

- Accuracy: 82.61%
- F1 Score: 85.05%
- Recall: 83.49%
- ROC-AUC: 82.32%

SVM performs similarly to Logistic Regression but with slightly improved accuracy and F1 score. The ROC-AUC score is good, though not as high as Random Forest, indicating its effectiveness in classification but with some room for improvement.

Decision Tree

- Accuracy: 77.72%
- F1 Score: 79.60%
- Recall: 73.39%
- ROC-AUC: 78.70%

The Decision Tree model has the lowest performance among the models in terms of accuracy and ROC-AUC. This might be due to its tendency to overfit or be less robust to the variability in the data.

Neural Network

- Accuracy: 80.43%
- F1 Score: 83.33%
- Recall: 82.57%
- ROC-AUC: 81.74%

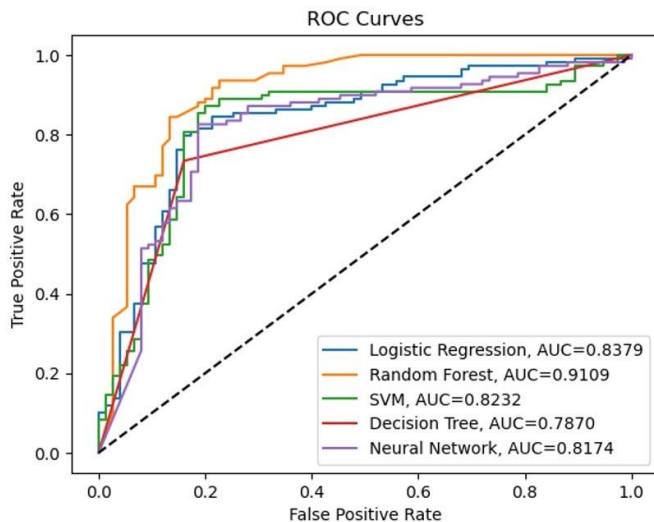
Neural Network shows competitive performance, with scores close to those of Logistic Regression and SVM. Its recall rate is commendable, indicating a good rate of correctly identifying positive cases, but its ROC-AUC suggests there's potential for improvement in differentiating between classes.

From the model results, we can summarize the following:

- **Best Overall Performance:** Random Forest stands out with the highest scores across all metrics.
- **Balance of Metrics:** Logistic Regression and SVM offer a good balance between accuracy, recall, and ROC-AUC.
- **Potential for Improvement:** Decision Tree and Neural Network, while reasonably effective, show potential areas for enhancement in classification accuracy and ROC-AUC scores.

G. Hyperparameter tuning

Hyperparameter tuning is the process of optimizing the hyperparameters of a machine learning model to achieve better performance. Hyperparameters are configuration settings external to the model that cannot be learned from the data but significantly impact the model's performance.



The goal of hyperparameter tuning is to find the optimal combination of hyperparameter values that maximizes the model's performance.

In our research, we have utilized the grid search technique, which involves defining a grid of hyperparameter values and training the model for all possible combinations.

Optimized Model Evaluation Metrics

1. Accuracy: 85.87%

- Indicates the proportion of total predictions (both heart disease and no heart disease) that were correct.
- High accuracy suggests the model is effective in distinguishing between the presence and absence of heart disease.

2. F1 Score: 87.96%

- A balance between precision and recall, indicating the model's strength in accurately identifying true positive cases while minimizing false positives.

An F1 score close to 88% is impressive, especially in medical diagnostics, where both false negatives and false positives have significant implications.

3. Recall: 87.16%

- Demonstrates the model's ability to identify true positive cases of heart disease.
- This high recall rate is crucial in medical scenarios to ensure that most patients with the condition are correctly identified.

4. ROC-AUC: 91.18%

- Measures the model's ability to distinguish between classes (heart disease vs. no heart disease) across different thresholds.
- An AUC of over 91% is excellent, indicating a high degree of separability achieved by the model.

ROC Curve Analysis

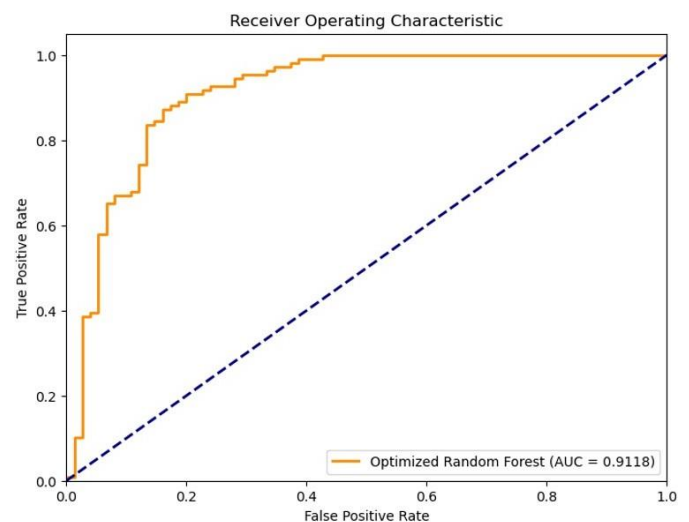
- The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate for different threshold values.
- The area under the curve (AUC) being approximately 91% reflects the model's high discriminative ability.
- The curve staying well above the diagonal line (representing a random guess) visually confirms the model's effectiveness.

H. Comparison of Initial and Optimized model

Comparing the performance metrics of the optimized Random Forest model with the initial model yields.

Accuracy

- Optimized Model: 85.87%
- Initial Model: 85.33%



- Analysis: There's a slight improvement in accuracy with the optimized model. This indicates a marginally better overall rate of correct predictions for both classes (heart disease and no heart disease).

F1 Score

- Optimized Model: 87.96%
- Initial Model: 87.20%
- Analysis: The F1 score, which balances precision and recall, is higher in the optimized model. This suggests a better balance in correctly identifying positive cases while minimizing false positives.

Recall

- Optimized Model: 87.16%
- Initial Model: 84.40%
- Analysis: There's a notable improvement in recall with the optimized model. This is significant in medical diagnostics, as a higher recall means the model is more effective at identifying true positive cases of heart disease.

ROC-AUC

- Optimized Model: 91.18%
- Initial Model: 91.09%
- Analysis: Both models exhibit an excellent ROC-AUC score, with a marginal improvement in the optimized model. This score reflects the model's ability to distinguish between the classes at various thresholds, and a higher score indicates better model performance.

VII. IMPACTS OF STUDY IN THE MEDICAL FIELD

The study on "Heart Disease Prediction using Machine Learning" has significant implications in the medical field, contributing to advancements in cardiovascular health interventions. The impacts of this research can be outlined as follows:

- Early Detection and Intervention:** The development of a robust machine learning-based predictive model enables early detection of heart diseases. Timely alerts provided by the model can facilitate proactive interventions, reducing the risk of severe cardiac events and ultimately improving patient outcomes.
- Algorithmic Evaluation:** The comprehensive exploration of various machine learning algorithms, including Logistic Regression, Decision Trees, Random Forests, Gradient Boosting Machines, Neural Networks, and Support Vector Machines, provides valuable insights into their efficacy in the nuanced realm of heart disease prediction. This evaluation aids in selecting the most suitable models for accurate predictions.
- Personalized Medicine:** Machine learning enables the analysis of individual patient data to tailor treatments based on genetic makeup, lifestyle, and other personalized factors. This approach enhances treatment effectiveness, reduces side effects, and improves overall patient outcomes.

VIII. CONCLUSION

The hyperparameter tuning of the Random Forest model has led to improvements across all key performance metrics, albeit the improvements are marginal. This suggests that the initial model was already performing quite well, but with tuning, its ability to predict heart disease accurately has been slightly enhanced. The increased recall rate is particularly important in a medical context, as it indicates a higher likelihood of correctly identifying patients with heart disease.

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